

Assignment 5: Unconsidered Metrics

Peripheral oxygen saturation (SpO₂) is defined as the percentage of haemoglobin in the blood that is saturated with oxygen. In the clinical setting, it is measured non-invasively using a pulse oximeter, which estimates how effectively oxygen is being transported throughout the body. SpO₂ is influenced by respiratory function, oxygen exchange in the lungs and the amount of oxygen being inhaled. In triage, nurses use SpO₂ as a crucial and rapid indicator of a patient's respiratory status and overall oxygenation. For example a patient may appear stable in the early stages of respiratory failure, by showing a normal respiratory rate or blood pressure. However, if the SpO₂ reading is low, this can indicate that vital organs are not receiving adequate oxygen. Monitoring SpO₂ helps nurses identify early signs of hypoxia, respiratory distress or other deteriorations that may not be noticeable by observation alone.

SpO₂ is used to assess the amount of oxygen delivery to the body's tissues and organs. The quantitative ranges to determine whether SpO₂ is within a normal or abnormal range are as followed:

1. (95% - 100%): The normal range and indicates excellent blood oxygen levels.
2. (92% - 94%): Borderline abnormal range. This indicates that there may be mild hypoxemia and could suggest the need for supplemental oxygen.
3. (92% or below): Hypoxemia. A reading within this range indicates that there is insufficient oxygen levels within the blood and may require further clinical assessment
4. (88% or below): Critical. Severe hypoxemia has occurred and suggests that the body's vital organs are not receiving enough oxygen.

Reference: <https://my.clevelandclinic.org/health/diagnostics/22447-blood-oxygen-level>